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Reference-Synchronized Low-Speed Velocity Measurement for a Round Inductosyn Using Digital Angle Output

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Abstract

Low-speed angular velocity extraction from round inductosyn feedback is affected not only by differentiation noise, but also by nonidealities of the digital sensing chain, including coarse–fine fusion, asynchronous transmission, reference synchronization, and evaluation delay. This paper presents a reference-synchronized measurement framework for low-speed velocity estimation using the fused 24-bit digital angle output of a round inductosyn demodulation board. An independent optical-encoder reference chain, a unified 1 ms evaluation timeline, DSP-side resampling, delay definition, and consistent error metrics are established to enable reproducible and fair comparison across estimators. On this basis, model-free methods, a kinematics-based Kalman filter, and a PMSM-model-based extended Kalman filter are implemented and evaluated under the same input and synchronization conventions. Simulations and hardware experiments, including open-loop multi-point step tests and closed-loop trapezoidal-motion tests, show that the proposed framework can clearly reveal the noise–latency trade-off of different estimators, while the EKF provides the most favorable balance in the tested conditions. The study contributes a measurement-oriented methodology for low-speed velocity evaluation of round inductosyn digital-output systems and related angular sensing chains.

1 Introduction

With the continuous advancement of servo control systems, increasing attention has been paid to the accuracy and dynamic response of rotor position and speed feedback [1, 2]. Common angular-feedback devices include optical encoders, capacitive sensors, and inductive angular sensors, as well as resolver-based interfaces widely used in servo systems [3–5]. For high-precision motion systems, sensing performance is influenced not only by nominal resolution, but also by metrological error sources and signal-conditioning quality [6]. Among inductive angular sensors, the round inductosyn remains attractive in aerospace, precision servo drives, and harsh industrial environments because it can maintain stable operation under high temperature, vibration, and strong electromagnetic interference [7–9]. Recent studies on round inductosyn systems have mainly focused on angle-measurement architectures, excitation and decoding circuits, and demodulation performance [10]. By contrast, low-speed velocity extraction from the digital angle output has been reported much less frequently, and a systematic measurement-oriented framework that clarifies the reference chain, synchronization, and evaluation metrics is still limited.

Existing rotor-speed estimation methods can generally be divided into difference-based and model-based approaches. The difference-based approach estimates speed directly from the position increment between adjacent sampling instants and is widely used because of its simplicity and low computational cost [11, 12]. In encoder-type sensing chains, edge-based state estimation and related timing strategies can further improve effective low-speed resolution [13]. Nevertheless, when the speed is very low, the angular increment within one sampling interval becomes small, so quantization effects and measurement noise can dominate the differentiated signal. As a result, difference-based estimates inherently face a trade-off between low-speed resolution and temporal responsiveness.

To mitigate differentiation noise, filtering and signal-tracking strategies are often introduced after position-to-speed conversion. Digital filtering can smooth the estimated speed and improve low-speed stability, but it usually introduces phase lag and degrades transient fidelity [14]. For analytical angular sensors, resolver-to-digital conversion and signal-tracking methods have also been developed to improve angular and speed extraction robustness, including excitation-aware RDC schemes, observer-based conversion, and phase-locked-loop-based extraction [15–19]. These methods can provide better smoothness and stronger noise immunity than direct differentiation alone, yet their practical performance still depends on sensor characteristics, bandwidth selection, parameter tuning, and sensing-chain latency. Therefore, the balance between low-speed precision and dynamic responsiveness remains a central issue in practical velocity measurement.

Model-based estimation provides another route for improving low-speed speed extraction. By combining system dynamics with measurement information, observer-based methods can suppress quantization effects and improve estimate consistency [13, 20, 21]. In related angular sensing systems, state-observer-based calibration and composite observer methods have also been reported for resolver signals, indicating that model-assisted processing can improve measurement reliability when the sensing chain is properly characterized [22, 23]. However, practical inductosyn-based low-speed velocity measurement involves additional nonidealities beyond the estimator itself. In particular, the sensing chain typically includes analog demodulation, coarse–fine fusion, and digital transmission, and the resulting position output may exhibit non-negligible latency relative to the control and acquisition timelines. Furthermore, meaningful comparison across different estimation methods requires an explicitly defined reference chain, synchronization strategy, delay convention, and unified error metrics. Without these elements, estimator improvements are difficult to interpret in a reproducible and measurement-oriented manner.

To address these issues, this paper develops a reference-synchronized framework for low-speed velocity measurement using the digital angle output of a round inductosyn, rather than presenting a standalone algorithm-only comparison. An independent circular grating encoder is introduced as the reference chain, and the time-alignment strategy, delay convention, and error metrics are explicitly defined on a unified evaluation timeline so that different estimators can be compared under reproducible conditions. Based on the fused digital position output of the inductosyn demodulation stage, model-free estimators are used as baselines, a kinematics-based Kalman filter is implemented as a linear model-assisted estimator, and a PMSM-model-based extended Kalman filter is further formulated to exploit the available current and position measurements within the same sensing chain. Simulations and hardware experiments are then conducted to evaluate estimator behavior under unified input, synchronization, and metric conventions. The results show that the proposed framework can clearly reveal the low-speed noise–latency trade-off across methods, while the extended Kalman filter provides the most favorable accuracy–responsiveness balance under the tested conditions.

The contributions of this work are threefold. First, a reference-synchronized evaluation framework is established for low-speed velocity measurement from round inductosyn digital angle output, in which the reference chain, unified time base, resampling strategy, delay definition, and error metrics are explicitly specified to enable fair and reproducible comparison. Second, the low-speed noise–latency limitations of model-free differentiation and filtering methods are analyzed on the same fused position input, and model-assisted estimators are implemented within the same digital sensing chain to isolate estimator effects from evaluation inconsistency. Third, open-loop and closed-loop experiments are carried out under unified synchronization and metric conventions, providing a measurement-oriented benchmark for low-speed velocity extraction in inductive angular sensing systems. The proposed framework is not limited to the tested round inductosyn hardware, but is also applicable to other asynchronous digital angle sensing chains, such as resolver/RDC systems and coarse–fine fused angular sensors.

2 Measurement system and reference framework

This section defines the measurement chain and reference protocol used for low-speed velocity evaluation of the round inductosyn digital-output system. By specifying the system architecture, digital-angle acquisition path, independent reference chain, synchronization strategy, and delay convention, the proposed framework establishes a reproducible basis for fair comparison across different velocity estimators.

2.1 System overview

This work develops a round inductosyn based low speed velocity measurement system with a unified reference framework for method comparison. As shown in Fig. 1, the system outputs the

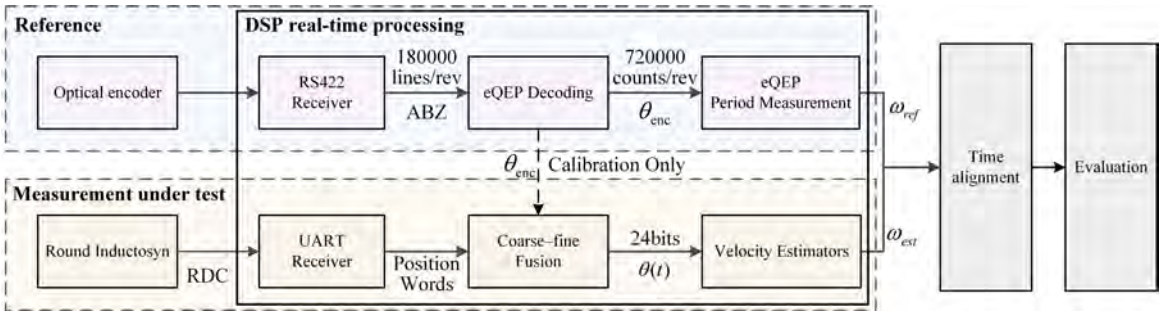

Figure 1: System block diagram of the round inductosyn based low speed velocity measurement and reference framework.

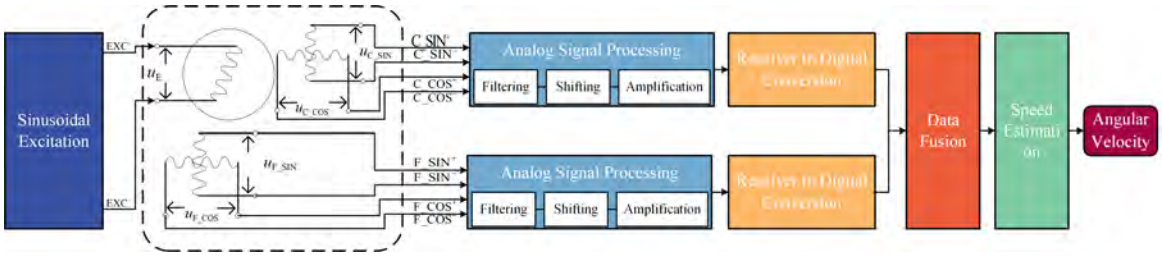

Figure 2: Signal processing and angular velocity estimation flow of the round inductosyn system.

fused inductosyn position $\theta(t)$ and the estimated velocity ω_{est}, while an independent reference chain provides the encoder position θ_{enc} and the reference velocity ω_{ref} for evaluation.

The reference chain employs a circular grating encoder, whose differential incremental channels A, B, and index Z (ABZ) are conditioned by an RS422 receiver and processed on a Texas Instruments TMS320F28335 digital signal processor (DSP) in real time. The on-chip enhanced quadrature encoder pulse (eQEP) module generates θ_{enc} from an encoder resolution of 180,000 lines per revolution, and four-times edge decoding yields an effective count resolution of 720,000 counts per revolution. Based on the decoded ABZ transitions, the eQEP period-measurement module provides the reference angular velocity ω_{ref} through edge-timing capture.

The measurement under test uses a round inductosyn together with a custom-built Resolver-to-Digital Conversion (RDC) demodulation board. The demodulation board outputs digital position words to the DSP through a universal asynchronous receiver-transmitter (UART) interface. A least-squares coarse–fine fusion is then executed on the DSP to reconstruct the final inductosyn position $\theta(t)$ with a 24-bit representation. The fused position $\theta(t)$ is used as the common input to all velocity estimators to ensure a consistent basis for comparison.

The drive unit consists of a permanent-magnet synchronous motor (PMSM) and a three-phase inverter driver board. The DSP executes the full real-time processing pipeline, including eQEP decoding and reference-velocity generation, coarse–fine fusion, and velocity estimation. The encoder position θ_{enc} is used for calibration and alignment only and is not directly fused into $\theta(t)$. The reference velocity ω_{ref} and the estimated velocity ω_{est} are subsequently synchronized by the time-alignment module and assessed using the unified evaluation conventions defined above.

2.2 Inductosyn signal model and data acquisition

The round inductosyn is a type of multi-pole inductive sensor that accurately reflects angular position by utilizing the principle that the electromagnetic induction signal between the planar windings of the stator and rotor varies periodically with displacement. Its excitation methods are categorized into single-phase excitation and bidirectional excitation, while signal demodulation methods include amplitude demodulation and phase demodulation. In this paper, a method based on amplitude demodulation using single-phase excitation on continuous windings and bidirectional output from segmented windings is adopted.

When a sinusoidal excitation signal of a certain frequency is applied to the continuous winding, the amplitude of the induced signal changes with the rotational displacement. The induced signal consists of sine and cosine components: when the signal of one phase is at zero, the other phase reaches its peak voltage. The two-phase output voltages provide an unambiguous representation of the angular position within one cycle. These analog signals are then processed through a signal

demodulation module involving filtering, phase shifting, and amplification, before being fed into the Resolver-to-Digital Conversion (RDC) module to detect the angular position. By fusing the angular data from both the coarse and fine channels and inputting it into a speed estimation module, an accurate angular velocity can be obtained.

The specific process is illustrated in Fig. 2. When the sinusoidal excitation given by (1) is applied to the continuous winding,

$$u(t) = U_m \sin \omega t \tag{1}$$

where U_m represents the amplitude of the sinusoidal excitation signal and ω represents its frequency, the sine and cosine induced EMF signals shown in (2) and (3), respectively, can be obtained.

$$e_c = -\frac{U_m}{k} \cos\theta \cos \omega t \tag{2}$$

$$e_s = -\frac{U_m}{k} \sin\theta \cos \omega t \tag{3}$$

Here, e_c and e_s represent the induced voltages on the cosine and sine windings, respectively; k is the fundamental voltage transfer coefficient, and t denotes time. Taking the round inductosyn with 180 pole pairs used in this study as an example, when a 200 Hz sinusoidal excitation is applied to the rotor winding, the induced signals in the sine and cosine channels, along with their envelope signals, are shown in Fig. 3. The first row depicts the sinusoidal excitation signal, while the second and third rows illustrate the induced signals and their envelopes in the sine and cosine channels, respectively.

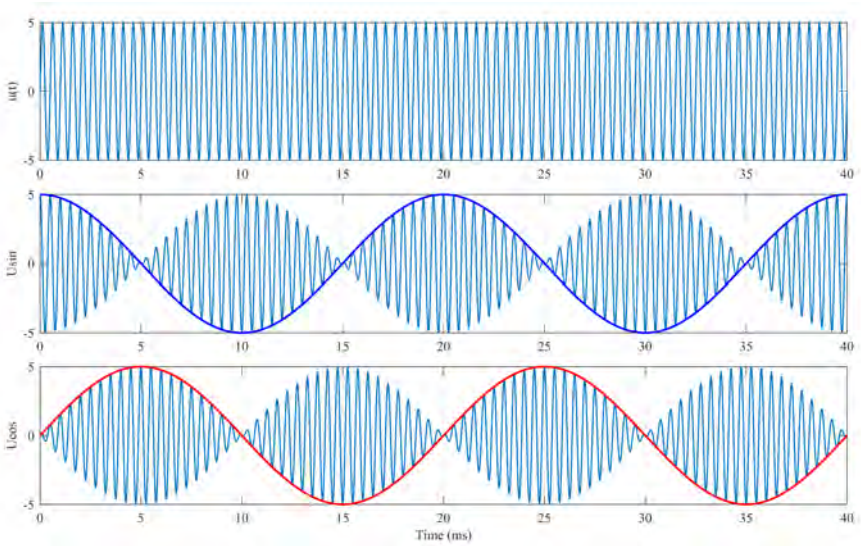

Figure 3: Induced signals and envelopes of sine and cosine channels under 200 Hz excitation.

In the proposed system, the DSP does not directly process the raw induced analog waveforms. The inductosyn signals are demodulated by an RDC-based stage, and the coarse–fine fusion is completed on the RDC demodulation board using a least-squares formulation. The board outputs a single fused 24-bit angular position word, which is transmitted to the TMS320F28335 through a UART link at a baud rate of 115200. After UART decoding, the DSP obtains a discrete-time position sequence $\theta[k]$ at sampling instants t_k, which serves as the unified position input to all angular velocity estimators in this paper.

Let $p[k]$ denote the received unsigned 24-bit code in the range $[0, 2^{24} - 1]$. The corresponding physical angle is defined as

$$\theta[k] = 2\pi \frac{p[k]}{2^{24}}, \tag{4}$$

which can be equivalently expressed in degrees as $\theta[k] \cdot 180/\pi$ when required. This definition provides a consistent mapping from the transmitted digital word to the physical angular position, enabling reproducible implementation of the subsequent velocity estimation and evaluation procedures.

To ensure data integrity, the UART frames are checked for completeness before converting the received bytes into the 24-bit position code. Invalid frames are discarded and the corresponding samples are marked as missing. The resulting position sequence is then processed by the real-time

velocity estimators on the DSP, while the reference and estimated velocities are aligned on a common evaluation timeline as described below.

2.3 Reference signals and synchronization

The optical encoder is decoded by the eQEP module with four-times edge decoding, yielding an effective resolution of $N_c = 720{,}000$ counts per revolution. The reference velocity is obtained from period measurement by accumulating a fixed number of encoder edges and measuring the elapsed time. Specifically, the eQEP capture logic updates the period measurement once every $N_u = 8$ encoder edges to improve stability at low speeds. Let $T_{\text{cap}}[m]$ denote the measured elapsed time for N_u edges at update index m. The reference angular velocity is computed as

$$\omega_{\text{ref}}[m] = \frac{2\pi}{N_c}\frac{N_u}{T_{\text{cap}}[m]}, \tag{5}$$

where $N_u = 8$ in this work. Compared with direct differentiation of position, period measurement based on edge timing improves stability in the low-speed regime because it relies on time-interval estimation rather than small position increments.

The reference and test signals are generated through different interfaces, namely eQEP for the encoder and UART for the inductosyn position word. To enable consistent evaluation in real time, the DSP establishes a unified evaluation timeline with a fixed grid of 1 ms, denoted as $t_k = kT_s$ with $T_s = 1$ ms. The fused inductosyn position $\theta[k]$ is obtained by UART decoding, and each velocity estimator outputs $\omega_{\text{est}}[k]$ on the same 1 ms grid. The eQEP period measurement provides ω_{ref} at nonuniform update instants determined by the N_u-edge accumulation. Therefore, the DSP resamples ω_{ref} onto the 1 ms grid using linear interpolation between adjacent eQEP updates to obtain $\omega_{\text{ref}}[k]$ for evaluation. Missing samples caused by invalid UART frames are marked and excluded from metric computation.

A delay term is defined to quantify the relative time shift between the estimated velocity and the reference velocity under the above synchronization conventions. The delay τ_d is obtained by maximizing the correlation between $\omega_{\text{est}}(t)$ and a time-shifted reference $\omega_{\text{ref}}(t-\tau)$ over the evaluated time interval,

$$\tau_d = \arg\max_{\tau} \operatorname{corr}\big(\omega_{\text{est}}(t), \omega_{\text{ref}}(t-\tau)\big). \tag{6}$$

A positive τ_d indicates that $\omega_{\text{est}}(t)$ lags behind $\omega_{\text{ref}}(t)$ under the adopted sign convention. All estimators are evaluated using the same position input $\theta[k]$, the same reference-velocity computation in equation (5), and the same DSP-side synchronization and delay conventions to ensure a fair comparison. The encoder-based reference chain is introduced as an independent comparison standard rather than as an auxiliary estimator input. Its edge-timing-based velocity generation provides a substantially more stable low-speed reference than direct differentiation of the inductosyn output. In addition, invalid UART frames are explicitly excluded, and the same interpolation, synchronization, and delay conventions are applied to all estimators. Therefore, the reported differences among methods primarily reflect estimator behavior under a common sensing chain, rather than inconsistencies in evaluation conditions.

3 Methods

3.1 Baseline Estimators for Low-Speed Digital-Angle Velocity Extraction

When the inductosyn is used as a position feedback element, the angular velocity is typically obtained by differentiating the position between two adjacent sampling points. However, at low speeds ($\leq 1°/\text{s}$), the presence of angle measurement errors and noise leads to a rapid increase in velocity estimation error. Although extending the sampling interval or adding a linear filter can help reduce the impact of noise, these methods introduce additional delays and reduce the system bandwidth.

3.1.1 Direct Differentiation-based Speed Estimation The principle of direct differentiation for velocity estimation is described by (7):

$$\hat{\omega}_k = \frac{\theta\left(t_k\right) - \theta\left(t_{k-1}\right)}{T_s} \tag{7}$$

where $\hat{\omega}_k$ represents the estimated angular velocity, $\theta(t_k)$ and $\theta(t_{k-1})$ are the measured angular positions at the current and previous sampling instants, respectively, and T_s is the sampling period. Considering the presence of measurement noise with a standard deviation Δ in position measurements, the worst-case scenario occurs when the position measurement errors at the two

sampling points have opposite signs and equal magnitude. Therefore, the peak velocity estimation error $\Delta\omega$ from the differentiation method is:

$$\Delta\omega = \frac{2\Delta}{\mathrm{T_s}} \tag{8}$$

The relative error e of the velocity estimation is defined as the ratio between the peak velocity estimation error $\Delta\omega$ and the true velocity ω, given by:

$$\mathrm{e} = \frac{\Delta\omega}{\omega} \times 100\% = \frac{2\Delta}{\mathrm{T_s}\omega} \times 100\% \tag{9}$$

Typically, for a 180 pole pair inductosyn, the fine channel has noise around $\pm 3''$, whereas the coarse channel exhibits noise of approximately $\pm 20''$. The error from the coarse channel is negligible in position data fusion, as it only serves to identify the cycle position of the fine channel signal. After fusing the coarse and fine channel data, the peak-to-peak error in the position signal is generally within $\pm 0.1''$. In engineering practice, the standard deviation of noise is usually considered as one-sixth of the peak value, giving a noise standard deviation (Δ) of $0.033''$. With a closed-loop velocity control cycle of $\mathrm{T_s} = 1$ ms, at a rotational speed of $\omega = 0.065°/\mathrm{s}$, the relative error e in velocity estimation using the direct differentiation method reaches 20.15%, which fails to satisfy the accuracy requirements for position feedback in the system.

3.1.2 Low-pass Filtering Speed Estimation The velocity signal obtained from direct differentiation contains a significant amount of high-frequency noise. A low-pass filter can be applied to preserve the low-frequency useful signal while attenuating the high-frequency noise. Commonly used low-pass filters include the first-order RC filter and the second-order Butterworth filter. The principle of the first-order RC filter is as follows:

$$\hat{\omega}_k = \alpha \cdot \omega_k + (1 - \alpha) \cdot \hat{\omega}_{k-1} \tag{10}$$

where α is the weighting coefficient of the filter. $\hat{\omega}_\mathrm{k}$ and $\hat{\omega}_{\mathrm{k}-1}$ represent the filtered velocity outputs at the current and previous time steps, respectively, while ω_k also denotes the raw velocity obtained from direct differentiation.

Compared with the first-order RC filter, the second-order Butterworth filter exhibits a steeper frequency attenuation characteristic, allowing it to suppress the high-frequency noise introduced by differentiation more effectively while maintaining a smooth system response. The principle of the second-order Butterworth filter is as follows:

$$\hat{\omega}_k = b_0\omega_k + b_1\omega_{k-1} + b_2\omega_{k-2} - a_0\hat{\omega}_{k-1} - a_1\hat{\omega}_{k-2} \tag{11}$$

$\hat{\omega}_k$, $\hat{\omega}_k - 1$ and $\hat{\omega}_k - 2$ represent the current and previous two outputs of the Butterworth filter, while ω_k, ω_{k-1} and ω_{k-2} are the current and previous two velocity values obtained from direct differentiation. The coefficients b_0, b_1, b_2, a_0, and a_1 are the filter parameters, which are determined based on the cutoff frequency (f_c) and sampling period (T_s) through a bilinear transformation.

3.1.3 Sliding Window Speed Estimation The sliding window method is a velocity estimation approach based on the fitting of multiple consecutive position samples. It is primarily used to overcome the large estimation errors associated with direct differentiation under low-speed or high-noise conditions. In this method, a fixed-length time window is used to perform first-order polynomial fitting on the position data to estimate the angular velocity at the current time.

Within each sliding window, the position data $(t_i; \theta_i)$ is approximated by a linear model:

$$\theta(t) = a \cdot t + b \tag{12}$$

where a is the slope of the fitted line and represents the estimated angular velocity. The least squares method is used to determine the coefficients a and b, and the slope a is computed as:

$$a = \frac{N\sum_{i=1}^{N} t_i\theta_i - \sum_{i=1}^{N} t_i \sum_{i=1}^{N} \theta_i}{N\sum_{i=1}^{N} t_i^2 - \left(\sum_{i=1}^{N} t_i\right)^2} \tag{13}$$

The estimated velocity at the current time is thus given by $\hat{\omega}_k = a$. By continuously shifting the window forward in time, real-time velocity estimation can be achieved with improved noise suppression compared to direct differentiation.

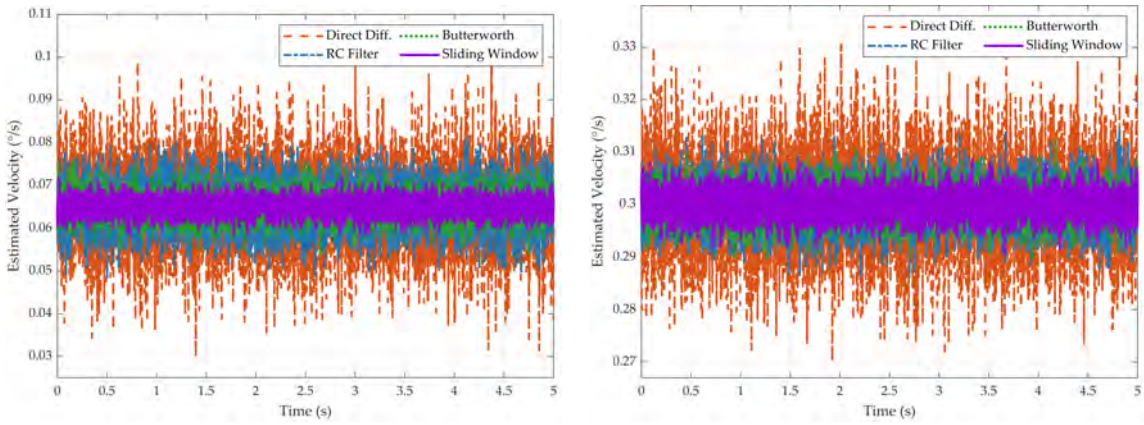

Figure 4: Velocity estimation results: (a) angular velocity = 0.065°/s. (b) angular velocity = 0.3°/s.

Table 1: Velocity Estimation Error Comparison at Low Speeds.

Angular Velocity (°/s)	Method	RMSE (°/s)	Max Error (°/s)	Relative Error (%)	Delay (ms)
0.065	Direct Differentiation	0.0131	0.0256	20.15	0
	RC Filter	0.0063	0.0125	9.69	2
	Butterworth Filter	0.0048	0.0098	7.38	5
	Sliding window	0.0037	0.0073	5.69	9
0.3	Direct Differentiation	0.0091	0.0183	6.1	0
	RC Filter	0.0042	0.0083	2.8	5
	Butterworth Filter	0.0031	0.0064	2.1	8
	Sliding window	0.0025	0.0052	1.7	10

3.2 Low-Speed Noise–Latency Characteristics of Model-Free Estimators

The angular position signal is generated based on a round inductosyn with a diameter of 12 cm and 180 pole pairs. Two typical low-speed scenarios 0.065°/s and 0.3°/s are selected as test cases. The sampling period is set to 1 ms, and the standard deviation of the angular signal noise is 1″. The sliding window method uses 10 sampling points per window, the RC filter employs a weighting coefficient of 0.1, and the cutoff frequency of the Butterworth filter is set to 10 Hz. The estimated velocity results are shown in Fig.4.

Under the low speed condition of 0.065°/s, the velocity signal estimated by the direct differentiation method contains significant high-frequency noise, resulting in large fluctuations and poor smoothness. Although the RC filter can effectively suppress part of the noise, noticeable jitter still remains. The second-order Butterworth filter shows better smoothing performance, and the sliding window method provides the most stable and noise-resistant result among the four, producing a smooth output that closely follows the true value.

Under the higher speed condition of 0.3°/s, the performance of all four algorithms improves. The direct differentiation method becomes acceptable, with reduced relative error. The RC and Butterworth filters both deliver smoother results, while the sliding window method continues to demonstrate the best overall performance in terms of noise suppression and output stability.

In summary, the model-free estimators exhibit different trade-offs between noise suppression and dynamic responsiveness in the low-speed regime. The sliding window method provides the strongest smoothing effect among the tested model-free approaches, whereas the Butterworth filter offers a more moderate compromise between fluctuation suppression and response delay. These methods therefore serve as meaningful baselines for the subsequent comparison, because they make the low-speed noise–latency trade-off explicit under the same position input and evaluation conventions.

3.3 Model-Assisted Estimators within the Digital Measurement Chain

To mitigate the low-speed noise–delay limitations observed in the above model-free estimators, two model-assisted estimators are implemented on the same fused inductosyn position input. A kinematics-based Kalman filter is first used as a linear baseline, and a PMSM-model-based extended Kalman filter is then formulated to exploit the available current and position

measurements within the same digital sensing chain. The purpose of introducing the EKF here is not to redefine the generic observer formulation itself, but to examine how model-assisted estimation performs when embedded in the practical round-inductosyn digital-output chain defined above under the synchronization conventions of the reference protocol.

3.3.1 Classical Kalman Filter For Speed Estimation To estimate the angular velocity of the motor, the following assumptions are made:

- The rotor undergoes uniformly accelerated motion over a short time interval, meaning the angular acceleration α is approximately constant during one sampling period.
- The rotor position θ is directly measurable.
- The system exhibits linear state transitions.

When the system is undergoing low-speed and steady motion, and the sampling frequency is high, the motion of the servo system can be simplified as a uniformly accelerated process. In this case, the kinematic equations of the system are:

$$\begin{cases} \theta_{k+1} = \theta_k + \omega_k \Delta t + \frac{1}{2}\alpha_k \Delta t^2 \\ \omega_{k+1} = \omega_k + \alpha_k \Delta t \\ \alpha_{k+1} = \alpha_k \end{cases} \tag{14}$$

its state equation and observation equation are:

$$\begin{cases} \hat{\mathbf{x}}_{\mathrm{k+1}} = \mathbf{A}\hat{\mathbf{x}}_{\mathrm{k}} + \mathbf{\Gamma}\mathrm{w}_{\mathrm{k}} \\ z_{\mathrm{k+1}} = \mathbf{H}\mathbf{x}_{\mathrm{k}} + \mathrm{v}_{\mathrm{k}} \end{cases} \tag{15}$$

where

$$\mathbf{A} = \begin{bmatrix} 1 & \Delta\mathrm{t} & \frac{1}{2}\Delta\mathrm{t}^2 \\ 0 & 1 & \Delta\mathrm{t} \\ 0 & 0 & 1 \end{bmatrix} \tag{16}$$

$$\Gamma = \begin{bmatrix} \frac{1}{2}\Delta\mathrm{t}^2 & \Delta\mathrm{t} & 1 \end{bmatrix}^{\mathrm{T}} \tag{17}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \tag{18}$$

$$\mathbf{x}_{\mathrm{k}} = \begin{bmatrix} \theta_{\mathrm{k}} & \omega_{\mathrm{k}} & \alpha_{\mathrm{k}} \end{bmatrix}^{\mathrm{T}} \tag{19}$$

$\boldsymbol{A}$ is the system state transition matrix, $\boldsymbol{H}$ is the observation matrix, and $\mathbf{\Gamma}$ is the system noise transition matrix. x_k is the state variable vector, and x_{k+1} is the estimated state vector. z_k represents the measured angular position. Δ_t is the sampling period, θ_k is the angular position, ω_k is the angular velocity, and α_k is the angular acceleration. w_k is the process noise due to modeling errors, and v_k is the observation noise. $\boldsymbol{Q}$ and $\boldsymbol{R}$ are the variances of w_k and v_k, respectively. The prediction stage of the Kalman filter includes state prediction and covariance prediction, given by the following equations.

State Prediction,Covariance Prediction:

$$\hat{\mathbf{x}}_{\mathrm{k}} = \mathbf{A}\mathbf{x}_{\mathrm{k-1}} \tag{20}$$

$$\hat{\mathbf{P}}_{\mathrm{k}} = \mathbf{A}\mathbf{P}_{\mathrm{k-1}}\mathbf{A}^{\mathrm{T}} + \hat{\mathbf{Q}} \tag{21}$$

After the prediction stage, the correction stage is performed, consisting of Kalman gain calculation, state update, and covariance update, as shown below:
Kalman Gain:

$$\mathbf{K}_{\mathrm{k}} = \hat{\mathbf{P}}_{\mathrm{k}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{P}_{\mathrm{k}}\mathbf{H}^{\mathrm{T}} + \hat{\mathbf{R}}\right)^{-1} \tag{22}$$

State Update:

$$\mathbf{x}_{\mathrm{k}} = \hat{\mathbf{x}}_{\mathrm{k}} + \mathbf{K}_{\mathrm{k}}\left(\mathbf{z}_{\mathrm{k}} - \mathbf{H}\hat{\mathbf{x}}_{\mathrm{k}}\right) \tag{23}$$

Covariance Update:

$$\mathbf{P}_{\mathrm{k}} = \left(\mathbf{I} - \mathbf{K}_{\mathrm{k}}\mathbf{H}\right)\hat{\mathbf{P}}_{\mathrm{k}} \tag{24}$$

3.3.2 Extended Kalman Filter for Speed Estimation Although the kinematics-based Kalman filter can effectively suppress low-speed differentiation noise, its motion model is still simplified and does not explicitly reflect the electromechanical coupling of the permanent-magnet synchronous motor (PMSM). To further improve the estimation accuracy under dynamic operating conditions, an extended Kalman filter (EKF) is constructed by incorporating the PMSM current and motion dynamics into the observer model.

In this work, the rotor position θ and angular velocity ω are both defined as *mechanical* quantities. Under the surface-mounted PMSM assumption, the nonlinear continuous-time model in the synchronous rotating d–q frame is written as

$$\begin{cases} \dot{i}_d = -\dfrac{R}{L_s} i_d + p\omega i_q + \dfrac{1}{L_s} u_d, \\ \dot{i}_q = -p\omega i_d - \dfrac{R}{L_s} i_q - \dfrac{p\psi_f}{L_s}\omega + \dfrac{1}{L_s} u_q, \\ \dot{\theta} = \omega, \\ \dot{\omega} = \dfrac{1}{J}\left(\dfrac{3}{2} p\psi_f i_q - B\omega - T_L\right), \end{cases} \tag{25}$$

where i_d and i_q are the stator currents in the synchronous rotating frame, u_d and u_q are the corresponding voltage inputs, R is the stator resistance, L_s is the stator inductance, ψ_f is the permanent-magnet flux linkage, p is the number of pole pairs, J is the equivalent inertia, B is the viscous damping coefficient, and T_L denotes the lumped load torque. In the practical EKF implementation, the uncertainty associated with T_L and other unmodeled disturbances is absorbed into the process-noise term.

The state vector, input vector, and measurement vector are defined as

$$x_k = \begin{bmatrix} i_{d,k} & i_{q,k} & \theta_k & \omega_k \end{bmatrix}^{\mathrm{T}}, \tag{26}$$

$$u_k = \begin{bmatrix} u_{d,k} & u_{q,k} \end{bmatrix}^{\mathrm{T}}, \tag{27}$$

$$y_k = \begin{bmatrix} i_{d,k} & i_{q,k} & \theta_k \end{bmatrix}^{\mathrm{T}}. \tag{28}$$

Using a forward-Euler discretization with sampling period T_s, the nonlinear discrete-time model is expressed as

$$x_{k+1} = x_k + T_s f(x_k, u_k) + w_k, \tag{29}$$

$$y_k = C x_k + v_k, \tag{30}$$

where

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \tag{31}$$

w_k is the process-noise vector and v_k is the measurement-noise vector. They are assumed to be mutually independent zero-mean white sequences with covariance matrices

$$w_k \sim \mathcal{N}(0, Q_k), \qquad v_k \sim \mathcal{N}(0, R_k), \tag{32}$$

where $Q_k \in \mathbb{R}^{4\times 4}$ and $R_k \in \mathbb{R}^{3\times 3}$.

Because the state equation is nonlinear, the EKF performs local first-order linearization around the previous updated estimate. The state prediction is

$$\hat{x}_{k|k-1} = \hat{x}_{k-1|k-1} + T_s f\big(\hat{x}_{k-1|k-1}, u_{k-1}\big), \tag{33}$$

and the corresponding predicted output is

$$\hat{y}_{k|k-1} = C\hat{x}_{k|k-1}. \tag{34}$$

The continuous-time Jacobian of the nonlinear state function is defined as

$$A_k = \left.\frac{\partial f(x,u)}{\partial x}\right|_{\hat{x}_{k-1|k-1},\, u_{k-1}}, \tag{35}$$

which yields

$$A_k = \begin{bmatrix} -\frac{R}{L_s} & p\hat{\omega}_{k-1|k-1} & 0 & p\hat{i}_{q,k-1|k-1} \\ -p\hat{\omega}_{k-1|k-1} & -\frac{R}{L_s} & 0 & -p\hat{i}_{d,k-1|k-1} - \frac{p\psi_f}{L_s} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{3p\psi_f}{2J} & 0 & -\frac{B}{J} \end{bmatrix}. \tag{36}$$

Under the same forward-Euler discretization, the discrete-time state-transition Jacobian is approximated by

$$F_k = I + T_s A_k. \tag{37}$$

The covariance prediction is then written as

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^{\mathrm{T}} + Q_k. \tag{38}$$

The EKF gain, state update, and covariance update are given by

$$K_k = P_{k|k-1} C^{\mathrm{T}} \left(C P_{k|k-1} C^{\mathrm{T}} + R_k \right)^{-1}, \tag{39}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \left(y_k - \hat{y}_{k|k-1} \right), \tag{40}$$

$$P_{k|k} = \left(I - K_k C \right) P_{k|k-1} \left(I - K_k C \right)^{\mathrm{T}} + K_k R_k K_k^{\mathrm{T}}. \tag{41}$$

Compared with the kinematics-based Kalman filter, the EKF explicitly incorporates the electromechanical coupling of the PMSM and uses both current measurements and the fused inductosyn position as inputs to the estimation process. Therefore, it is more suitable for the practical digital sensing chain considered in this work, especially when low-speed quantization noise, dynamic current variation, and model uncertainty jointly affect the velocity estimation accuracy.

4 Simulation and Experiment

This section evaluates the proposed reference-synchronized measurement framework and compares different estimators under unified input, synchronization, and metric conventions. Simulations are first used to illustrate estimator behavior under controlled conditions, after which open-loop and closed-loop experiments are conducted on the physical platform to verify the practical low-speed measurement performance of different estimators within the same sensing chain. Both loops adopted conventional proportional–integral (PI) control, and the model parameters are listed in Table 2.

Table 2: Model Parameters.

Parameters	Symbol	Values	Unit
Number of pole pairs	Pn	60	-
Phase resistance	R	4.4	Ω
Phase inductance	Ls	0.025	H
Magnetic flux linkage	ψ_f	0.175	Wb
Moment of inertia	Jm	0.0008	kg·m^2
Damping coefficient	Bm	0	Nm·s/rad

4.1 Simulation Results

The motor is operated in speed mode. Two representative speed commands are applied, including a step command of 20 °/s and a sinusoidal command with an amplitude of 100 °/s. The corresponding estimation results are shown in Fig. 5 and Fig. 6, including the estimated angular velocity and position as well as their estimation errors.

The KF initialization used in the simulation is defined as follows:

$$P_0 = \begin{bmatrix} 0.04 & 0 & 0 \\ 0 & 0.01 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \; Q = 1e-2, \; R = 5e-3 \tag{42}$$

The EKF initialization and covariance matrices used in the simulation are defined as follows:

$$\mathbf{P}_0 = \mathrm{diag}(0.01,\, 1,\, 0.4,\, 0.25) \tag{43}$$

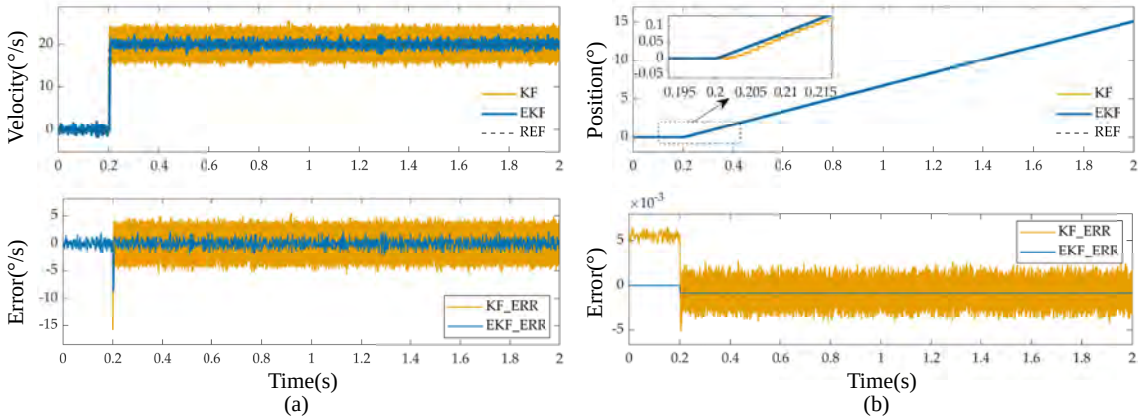

Figure 5: Step input response: (a) speed estimation and estimation error. (b) position estimation and estimation error.

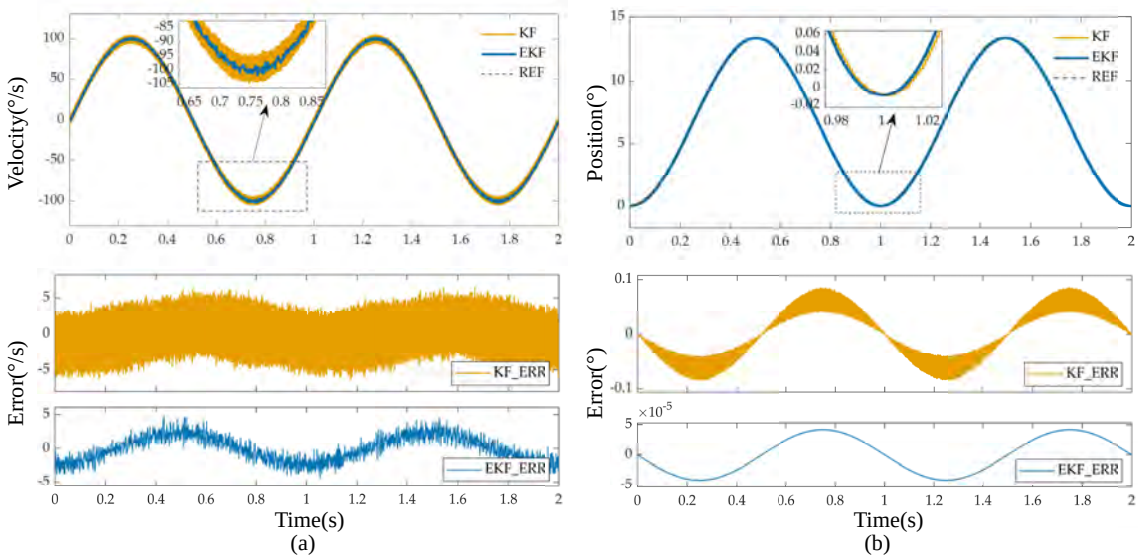

Figure 6: Sinusoidal input response: (a) speed estimation and estimation error. (b) position estimation and estimation error.

Table 3: Estimated simulation metrics for step and sinusoidal inputs.

Method	Metrics	Step input	Sinusoidal input
KF	RMSE_ω	2.6 °/s	3.5 °/s
	$E_{\max,\omega}$	15.0 °/s	6.0 °/s
	RMSE_θ	1.5×10^{-3} °	0.057 °
	$E_{\max,\theta}$	5.0×10^{-3} °	0.090 °
	t_s	0.03 s	–
EKF	RMSE_ω	1.0 °/s	2.1 °/s
	$E_{\max,\omega}$	9.0 °/s	4.0 °/s
	RMSE_θ	5.0×10^{-5} °	3.2×10^{-5} °
	$E_{\max,\theta}$	1.0×10^{-4} °	4.5×10^{-5} °
	t_s	0.02 s	–

$$\mathbf{Q} = \mathrm{diag}\big(5 \times 10^{-3},\, 5 \times 10^{-2},\, 1 \times 10^{-6},\, 2 \times 10^{-2}\big) \tag{44}$$

$$\mathbf{R} = \mathrm{diag}\big(4 \times 10^{-4},\, 4 \times 10^{-4},\, 5 \times 10^{-3}\big) \tag{45}$$

As shown in Fig. 5, both observers converge to stable tracking after the speed step at $t = 0.2$ s. Nevertheless, the KF exhibits a wider error envelope and a larger transient peak, whereas the EKF

provides tighter error containment. This is quantitatively confirmed by Table 3. For the step input, the EKF reduces the velocity RMSE from 2.6 to 1.0 °/s and the peak velocity error from 15.0 to 9.0 °/s compared with the KF. The improvement is more pronounced in the position channel: RMSE_θ decreases from 1.5×10^{-3} to 5.0×10^{-5} °, and $E_{\max,\theta}$ decreases from 5.0×10^{-3} to 1.0×10^{-4} °, corresponding to approximately 30× and 50× reductions, respectively. The settling time also decreases from 0.03 s (KF) to 0.02 s (EKF).

As shown in Fig. 6, the sinusoidal command introduces continuously time-varying speed, which places a more stringent requirement on dynamic tracking. In this case, the KF suffers from increased deviations in both velocity and position errors, while the EKF maintains a more compact error distribution over the entire cycle.

As listed in Table 3, the EKF reduces the velocity RMSE from 3.5 to 2.1 °/s and the peak velocity error from 6.0 to 4.0 °/s. More importantly, the EKF achieves substantially improved position consistency: the KF produces $\mathrm{RMSE}_\theta = 0.057$ ° and $E_{\max,\theta} = 0.090$ °, whereas the EKF remains at $\mathrm{RMSE}_\theta = 3.2 \times 10^{-5}$ ° and $E_{\max,\theta} = 4.5 \times 10^{-5}$ °. This indicates that the EKF provides a significantly more reliable position reconstruction under time-varying motion, which is critical because position consistency directly affects the derived velocity and the subsequent closed-loop evaluation.

In summary, the simulations indicate that both observers can achieve stable tracking under the adopted model and noise settings, whereas the EKF yields tighter velocity and position estimates than the linear KF baseline. These results mainly serve to illustrate estimator behavior under controlled conditions. The measurement relevance of the comparison is further assessed through the hardware experiments reported below, where the same synchronization and evaluation conventions are retained.

4.2 Experiment Results

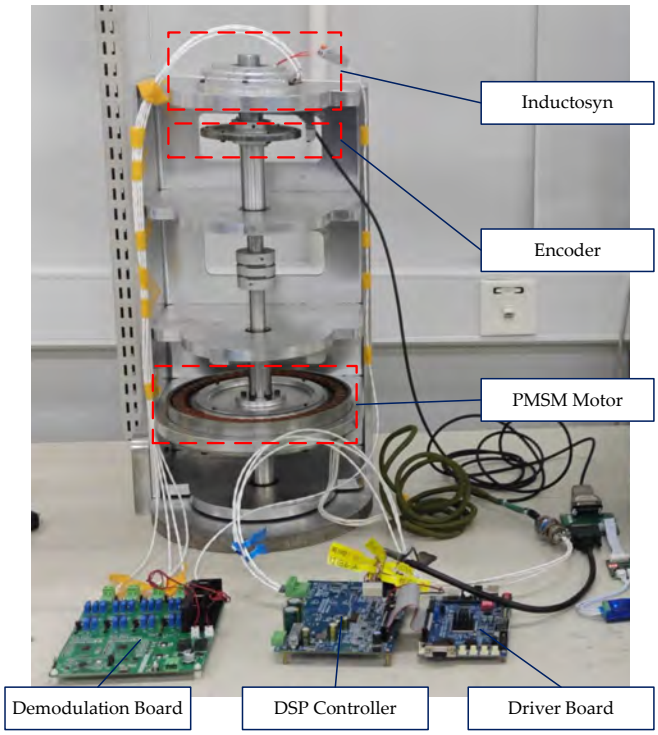

Figure 7: Experimental setup of speed estimation and closed-loop testing.

The experimental setup used for speed estimation and closed-loop testing is shown in Fig. 7. From top to bottom, the main components are the round inductosyn (Changzhou Weinengda Precision Machinery Co., Ltd, JGX210), a 24-bit circular grating encoder (RENISHAW, RSM20USB115), and the permanent-magnet synchronous motor (CETC Robotics and the 21st Research Institute, J210LWX001). The inductosyn sensing signals are connected to a custom-built position demodulation board, which converts the analog sensing signals into digital position words and transmits them to the DSP control board through a UART interface. The DSP board then generates space-vector pulse-width modulation (SVPWM) signals, which are applied to the three-phase inverter circuit on the driver board to actuate the motor. The position and speed

signals provided by the circular grating encoder are used as reference values. In the open-loop test, these signals are also used as feedback for the speed loop, whereas in the closed-loop test they serve only for comparison with the estimated position and velocity and are not used as speed feedback. During the experiments, the circular grating encoder was configured with 180,000 lines, and after quadrature decoding by the DSP controller, the effective resolution reached 720,000 counts per revolution.

4.2.1 Encoder-closed-loop actuation with open-loop estimator evaluation First, open-loop speed step tests were conducted at reference speeds of 0.065°/s, 0.3°/s, 0.5°/s, and 1.0°/s. The dynamic estimation performance of the Butterworth filter, the sliding window method, the Kalman filter (KF), and the extended Kalman filter (EKF) was evaluated using the same fused inductosyn position input, the same encoder-based reference chain, and the same synchronization and delay conventions defined above. Therefore, the reported differences reflect estimator characteristics under a common measurement protocol rather than inconsistencies in input or evaluation conditions. The corresponding speed waveforms are shown in Fig. 8.

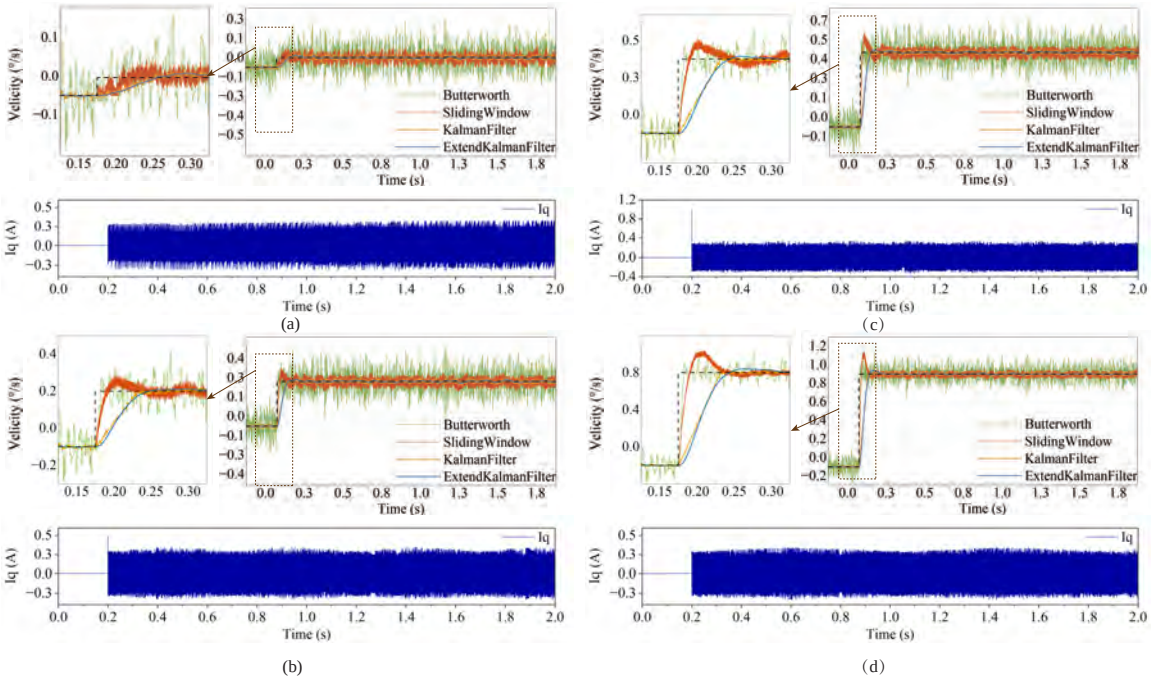

Figure 8: Open-loop speed estimation results at different reference speeds: (a) 0.065 °/s. (b) 0.3 °/s. (c) 0.5 °/s. (d) 1.0 °/s.

Table 4: Summary of step-response velocity metrics and corresponding i_q statistics at four commanded speeds.

Commanded speed	Method	Velocity step metrics							Q-axis current metrics		
		τ_d (ms)	RMSE (°/s)	P95 (°/s)	$E_{\max}$ (°/s)	t_r (ms)	M_p (%)	t_s (s)	$\lvert I_q\rvert_{\max}$ (A)	J_q (A s)	σ_{I_q} (A)
0.065 °/s	Butterworth	317.3	0.805	1.549	2.630	**4.0**	409.0	–	0.313	0.056	0.109
	Sliding window	29.6	0.205	0.417	0.707	14.0	125.2	–			
	Kalman filter	8.2	0.087	0.164	0.751	65.0	45.5	–			
	Extended Kalman filter	**6.2**	**0.042**	**0.082**	**0.649**	56.0	**24.4**	–			
0.3 °/s	Butterworth	2.1	0.773	1.427	3.210	**4.0**	73.0	–	0.349	0.059	0.114
	Sliding window	1.6	0.169	0.323	3.005	10.5	29.8	–			
	Kalman filter	**1.0**	0.086	0.167	3.060	46.0	9.1	–			
	Extended Kalman filter	2.1	**0.044**	**0.093**	**2.998**	44.0	**4.5**	**0.147**			
0.5 °/s	Butterworth	2.1	0.775	1.449	5.210	**5.0**	46.7	–	0.500	0.058	0.111
	Sliding window	1.1	0.166	0.315	5.005	10.5	28.2	–			
	Kalman filter	**1.0**	0.085	0.158	5.060	45.0	5.2	–			
	Extended Kalman filter	2.1	**0.042**	**0.080**	**4.997**	44.0	**3.8**	**0.208**			
1.0 °/s	Butterworth	2.1	0.764	1.447	10.210	**5.0**	23.8	–	1.000	0.056	0.111
	Sliding window	**0.7**	0.171	0.329	10.005	11.0	24.3	–			
	Kalman filter	1.0	0.083	0.155	10.060	45.0	**2.6**	**0.056**			
	Extended Kalman filter	2.1	**0.036**	**0.066**	**9.997**	43.0	3.9	0.129			

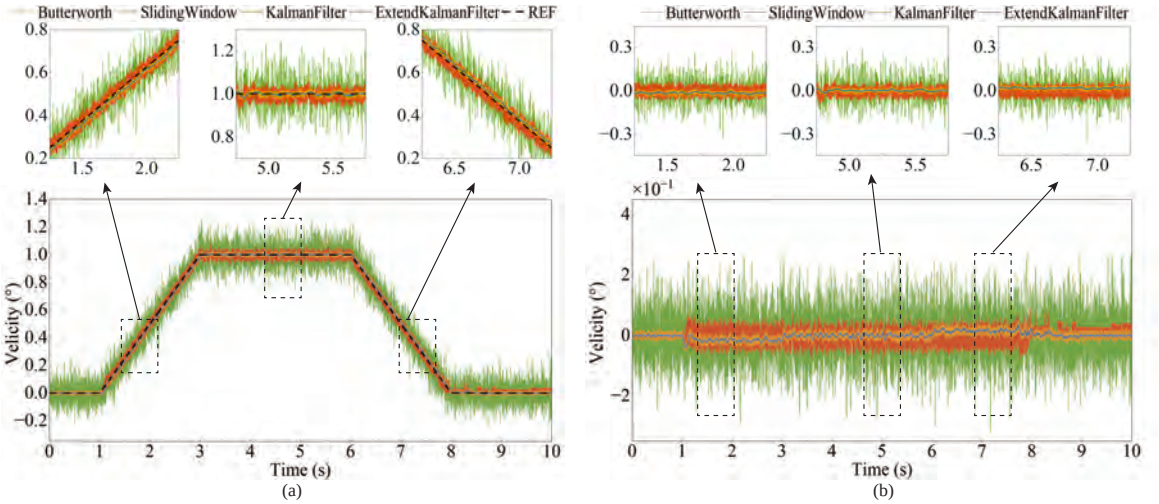

Figure 9: Closed-loop speed tracking and estimation error: (a) uniformly accelerated motion tracking; (b) speed error.

Table 4 summarizes the quantitative step-response metrics. At the lowest speed of 0.065°/s, the Butterworth and sliding window methods respond rapidly (rise times of only a few to tens of milliseconds), yet their estimation quality deteriorates markedly, as reflected by the large steady-state errors (RMSE and P95) and pronounced overshoot. In particular, the Butterworth output exhibits a very large delay ($\tau_d = 317.3$ ms) and the largest steady-state dispersion (RMSE= 0.805°/s and P95= 1.549°/s), indicating that aggressive low-pass filtering suppresses high-frequency noise at the cost of substantial phase lag and loss of low-speed fidelity. The sliding window method significantly reduces the delay ($\tau_d = 29.6$ ms) and improves the steady accuracy relative to Butterworth, but it still yields considerable overshoot and transient deviations, which is consistent with the inherent trade-off between differentiation-like responsiveness and noise amplification at ultra-low speeds.

In contrast, the model-based estimators remain substantially more stable in the same regime. KF and EKF introduce longer rise times (on the order of 50–65 ms) due to the estimator dynamics and smoothing effect, but they provide markedly lower steady-state errors. EKF achieves the smallest RMSE and P95 at 0.065°/s (0.042°/s and 0.082°/s, respectively) while also maintaining the smallest overshoot among the four methods, demonstrating improved robustness to quantization noise and measurement fluctuations under very low-speed operation.

As the operating speed increases from 0.3°/s to 1.0°/s, the delay of the Butterworth and sliding window methods becomes small (approximately 1–2 ms), indicating that their phase lag is less critical when the signal-to-noise ratio improves. However, their accuracy advantage does not scale accordingly: Butterworth remains the least accurate across all speeds, and the sliding window method continues to show evident overshoot, while KF and EKF keep overshoot at a low level. EKF consistently yields the lowest RMSE among all compared methods for each commanded speed, confirming that incorporating system dynamics and measurement statistics provides a more favorable balance between transient tracking and noise suppression.

Furthermore, the i_q statistics reported in Table 4 provide a consistent disturbance context for interpreting the transient estimation behavior. The transient current demand characterized by $|I_q|_{\max}$ increases with the commanded speed, which coincides with the growth of the transient error bound $E_{\max}$. Meanwhile, the steady-state current ripple level σ_{I_q} remains comparable across the tested conditions, suggesting that the observed RMSE and P95 differences primarily originate from the estimator characteristics rather than from changes in the disturbance baseline. Overall, these results verify that the EKF offers the best comprehensive performance for low-speed step tracking, achieving the smallest steady-state errors while avoiding the excessive overshoot observed in model-free methods.

4.2.2 Closed-loop speed response test To further compare the performance of different velocity estimation strategies under realistic servo operation, closed-loop experiments were conducted in which each estimation output was used as the feedback signal of the speed loop. The command followed a trapezoidal speed profile, and the reference speed was provided by the circular grating encoder. The measured responses in Fig. 9 indicate that all methods can track the commanded

trajectory, while their estimation errors differ substantially in both the cruise segment and the acceleration and deceleration transients. Quantitative results are summarized in Table 5.

In the cruise window from 3 s to 6 s at 1.0°/s, Butterworth exhibits the largest error spread, with $\mathrm{ERMS}_{3-6} = 0.079°/\mathrm{s}$, $\mathrm{P95}_{3-6} = 0.154°/\mathrm{s}$, and a peak error $E_{\max,3-6} = 0.257°/\mathrm{s}$. The sliding window method reduces the cruise dispersion ($\mathrm{ERMS}_{3-6} = 0.019°/\mathrm{s}$, $\mathrm{P95}_{3-6} = 0.036°/\mathrm{s}$) but still shows a noticeable peak ($E_{\max,3-6} = 0.076°/\mathrm{s}$). The conventional Kalman filter further improves steady performance with $\mathrm{ERMS}_{3-6} = 0.009°/\mathrm{s}$ and $E_{\max,3-6} = 0.031°/\mathrm{s}$. The EKF achieves the most concentrated cruise error distribution, yielding the lowest $\mathrm{ERMS}_{3-6} = 0.005°/\mathrm{s}$, $\mathrm{ESTD}_{3-6} = 0.005°/\mathrm{s}$, $\mathrm{P95}_{3-6} = 0.009°/\mathrm{s}$, and $E_{\max,3-6} = 0.016°/\mathrm{s}$.

Table 5: Closed-loop trapezoidal speed test: error metrics in cruise and dynamic segments.

Method	ERMS_{3-6}	ESTD_{3-6}	$E_{\max,3-6}$	$\mathrm{P95}_{3-6}$	ERMS_{1-8}	$E_{\max,1-8}$	$E_{\max,2.8-3.5}$	$E_{\max,5.8-6.5}$
Butterworth	0.079	0.079	0.257	0.154	0.078	0.305	0.253	0.281
Sliding window	0.019	0.018	0.076	0.036	0.018	0.084	0.065	0.064
Kalman filter	0.009	0.009	0.031	0.018	0.014	0.049	0.038	0.037
Extended Kalman Filter	**0.005**	**0.005**	**0.016**	**0.009**	**0.012**	**0.037**	**0.023**	**0.024**

Across the full dynamic window from 1 s to 8 s, the same ranking is observed. The EKF attains the smallest overall error ($\mathrm{ERMS}_{1-8} = 0.012°/\mathrm{s}$ and $E_{\max,1-8} = 0.037°/\mathrm{s}$), while the Kalman filter yields slightly larger residuals ($\mathrm{ERMS}_{1-8} = 0.014°/\mathrm{s}$ and $E_{\max,1-8} = 0.049°/\mathrm{s}$). Butterworth and the sliding window method present substantially larger peaks over the full trajectory. To specifically characterize the corner transients of the trapezoidal profile, the peak errors in 2.8–3.5 s and 5.8–6.5 s were evaluated. The EKF again provides the lowest transient peaks, with $E_{\max,2.8-3.5} = 0.023°/\mathrm{s}$ and $E_{\max,5.8-6.5} = 0.024°/\mathrm{s}$, indicating improved error containment during acceleration-to-cruise and cruise-to-deceleration transitions.

Overall, the results demonstrate that the EKF consistently provides the best closed-loop performance in terms of both steady cruise accuracy and transient error containment. More importantly, the proposed evaluation framework is able to distinguish not only steady-state error levels but also transient error behavior under low-speed trapezoidal motion, which is essential for practical feedback characterization of inductive angular sensing systems.

5 Conclusion

This paper investigated low-speed angular velocity measurement using the fused digital angle output of a round inductosyn and established a reference-synchronized measurement framework for reproducible estimator evaluation. The framework explicitly defines the independent encoder reference chain, DSP-side synchronization procedure, unified 1 ms evaluation timeline, delay convention, and consistent error metrics, thereby enabling fair comparison across different velocity estimators under the same sensing-chain conditions.

Within this framework, model-free estimators, a kinematics-based KF, and a PMSM-model-based EKF were implemented and compared using the same fused position input. Simulation and hardware experiments show that the framework can clearly reveal the low-speed noise–latency trade-off of different estimators, while the EKF provides the most favorable balance among the tested methods under both open-loop and closed-loop conditions.

The main contribution of this work lies in the measurement methodology and evaluation protocol rather than in a generic reformulation of Kalman filtering itself. The presented approach provides a practical benchmark for low-speed velocity extraction from round-inductosyn digital-output systems and is also applicable to other asynchronous digital angle sensing chains, including resolver/RDC systems and coarse–fine fused angular sensors.

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Author contributions

Methodology, J.W. (Jian Wang); software, C.Z.; formal analysis, Y.Z.; investigation, J.W. (Jian Wang); writing—original draft preparation, J.W. (Jian Wang); writing—review and editing, J.W. (Jianyuan Wang); supervision, J.W. (Jianyuan Wang); project administration, J.W. (Jianyuan

Wang); funding acquisition, J.W. (Jian Wang) and J.C. All authors have read and agreed to the published version of the manuscript.

Data availability

All the relevant data supporting the findings of this study are available within the paper

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